POWERTREAD
TEAM AUTO-X

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Abstract

PowerTread addresses urban energy crises and waste challenges by converting wasted human energy from footsteps into electrical power using piezoelectric transducers.

This innovative technology provides year-round energy generation, optimizing waste collection and offering valuable crowd density data for urban planning.

With applications in government, private sectors, religious centers, real estate, education, and recreation, PowerTread stands out in the market, meeting the increasing demand for sustainable solutions.

Our dedicated team, backed by expertise and partnerships, is committed to shaping a greener, more efficient future.

Methods and Materials

Basic Materials:
1. Piezoelectric Transducers: Square-shaped devices installed in high-traffic areas.
2. Regulation and Rectification circuit
3. Rechargeable Batteries: Store electrical energy for consistent use.

Methodology:
1. Piezoelectric Conversion: Converts applied forces from pedestrian footsteps into electric voltage.
2. Circuitry: Voltage boost, rectification and regulation.
3. Energy Storage: Batteries store generated electricity, ensuring consistent power supply.
4. Loads: Battery voltage converter to an A.C through inverter technology or directly connected to DC loads
5. Density Data collection

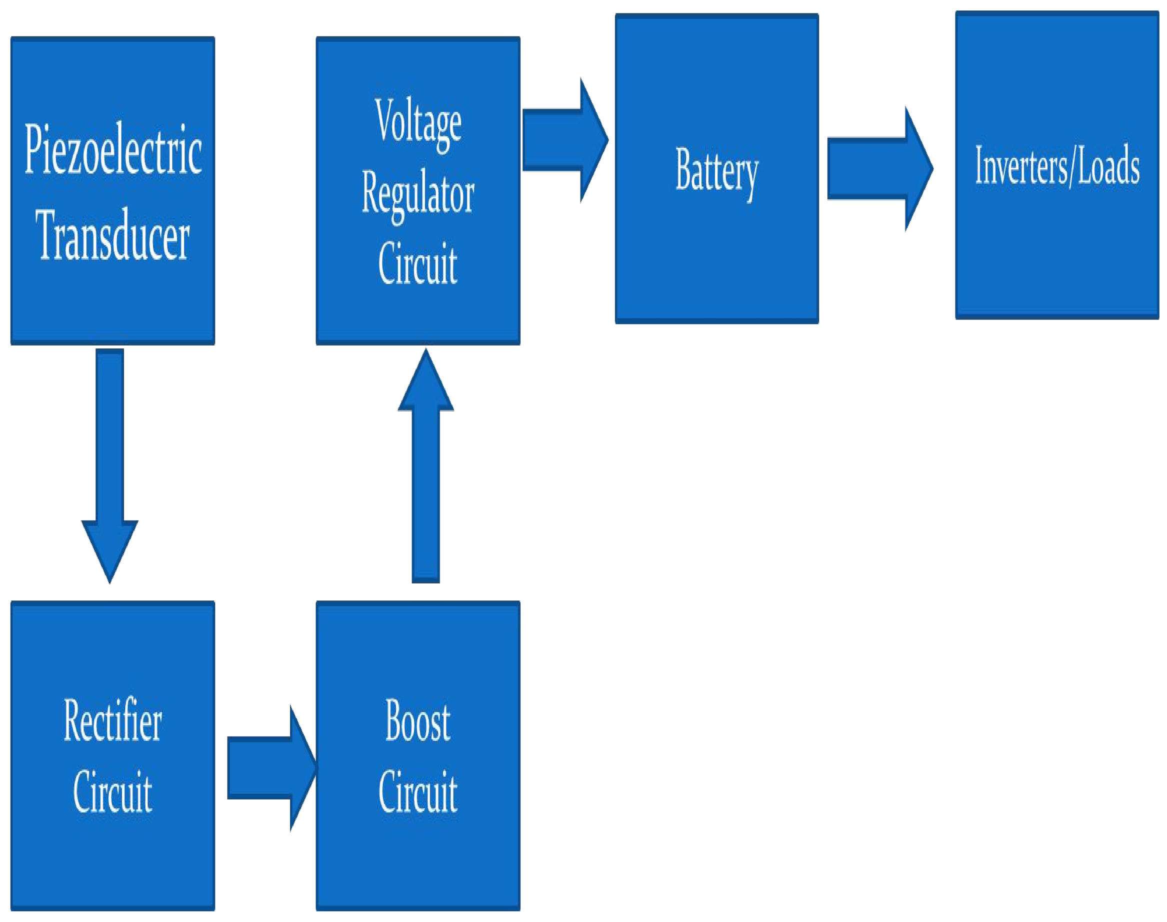


Chart 1. Methodology

Introduction

We focused on transforming urban spaces into sustainable energy hubs. By harnessing the mechanical energy from pedestrian footsteps through piezoelectric technology, PowerTread not only addresses the looming urban energy crisis but also revolutionizes waste management.

This project seeks to create a greener and more sustainable future by combining innovative technology, efficient waste collection systems, and community engagement initiatives.

PowerTread emerges as a beacon of urban sustainability, offering a holistic solution to the pressing challenges of energy conservation and waste management.

1. Government Entities
2. Private Sector
3. Religious Centers
4. Real Estate Developers
5. Educational Institutions
6. Recreation/Entertainment Venues

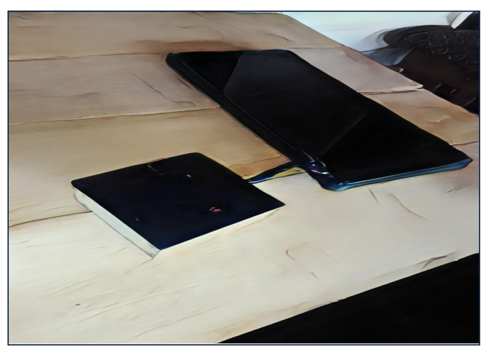


Figure 1. PowerTread.

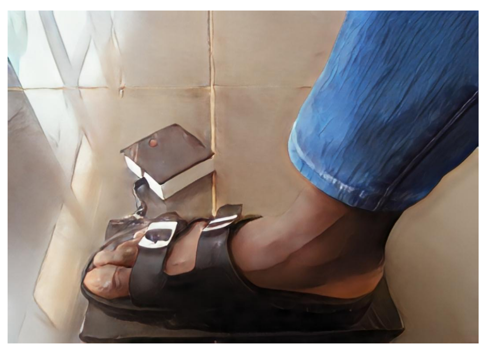


Figure 2. FootStep to Power.

Results

Achieving a total power of 720mW from 36 piezo transducers, each contributing 20mW, showcases the efficacy of PowerTread.

The first prototype's dimensions of 14cm by 30cm demonstrate the potential for scalability.

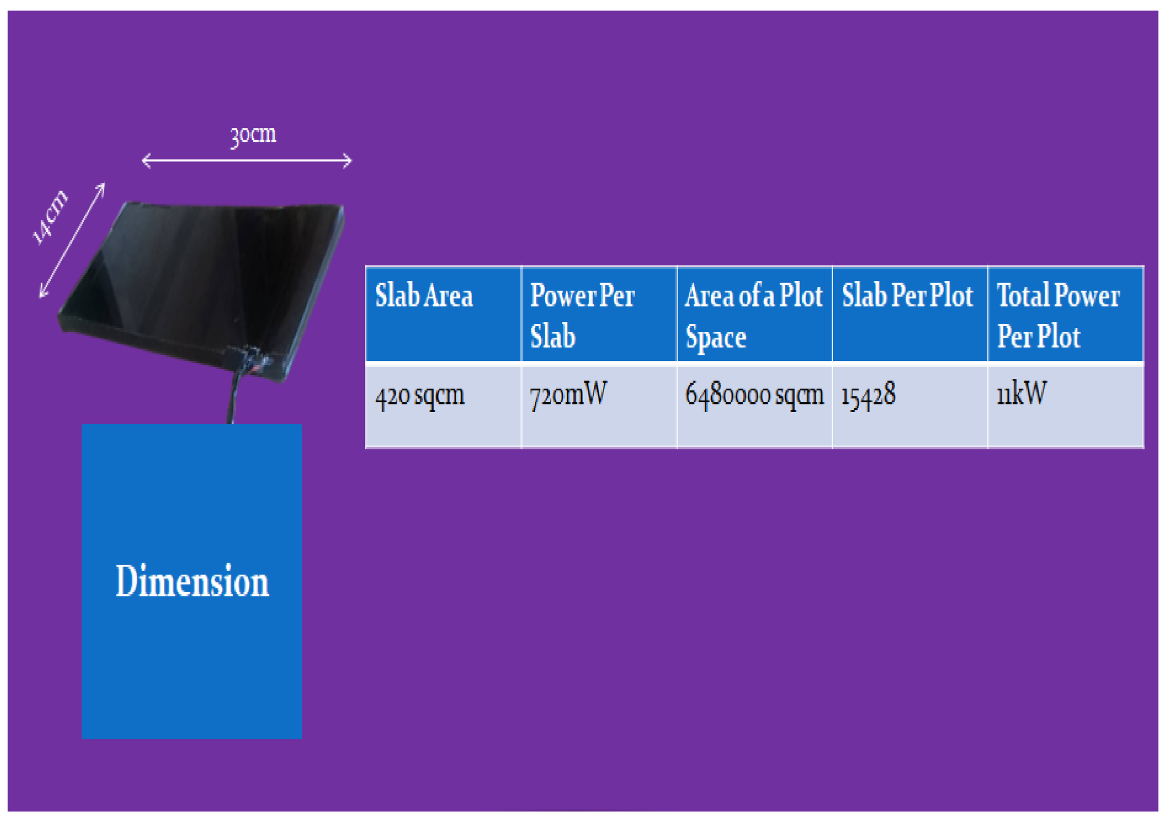
Extrapolating to a larger scale, with an array of PowerTreads installed in a plot of land, the projected power output of 11kW is indeed a significant leap towards sustainable energy solutions.

This success reinforces PowerTread's promise as a powerful and scalable technology in the realm of urban energy generation.

Power Per Transducer	Total Number of Transducer Per Slab	Total Power Per Slab
20mW	36	720mW

Number of Slabs	Total Power
50	36000 mW
70	50400 mW
100	72000 mW
150	108000 mW

Tables: Result



Slab Area	Power Per Slab	Area of a Plot Space	Slab Per Plot	Total Power Per Plot
420 sqcm	720mW	6480000 sqcm	15428	11kW

Discussion

PowerTread innovatively captures energy from footsteps using piezoelectric transducers, providing a reliable year-round solution to the urban energy crisis.

This groundbreaking technology not only generates electricity but also optimizes waste management and crowd monitoring.

With a scalable prototype projecting 11kW on a larger scale, PowerTread meets the demand for sustainable urban solutions. I

ts unique approach, combining energy efficiency with data-driven urban planning, positions it as a promising solution for government, private, religious, educational, and recreational sectors.

Conclusions

In conclusion, PowerTread stands as a transformative solution to urban energy challenges offering a unique blend of sustainable energy generation, waste management optimization, and crowd monitoring.

With a committed team and demonstrated scalability, PowerTread aligns seamlessly with market demands and presents a promising future for diverse sectors.

The technology's year-round efficiency and simultaneous benefits position it as a beacon for a greener, more resilient urban landscape, promising lasting positive impacts on energy consumption and environmental sustainability.

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